Fiscal Year:	FY 2016	**Task Last Updated:**	FY 01/18/2017
PI Name:	Hogan, Harry Ph.D.		
Project Title:	Pre-Treatment Approaches for Improving the Response of Bone to Disuse and Recovery		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Animal Biology: Vertebrate		
Space Biology Cross-Element Discipline:	(1) Musculoskeletal Biology		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	77843-3123	**Congressional District:**	17
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	09/01/2013	**End Date:**	08/31/2017
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	3	**No. of Master' Degrees:**	1
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 8/31/2017 per NSSC information (Ed., 6/3/16) NOTE: Period of performance dates corrected per NSSC information (Ed., 9/10/2013)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Allen, Matthew Ph.D. (Indiana University) Bloomfield, Susan (Texas A&M University) Fluckey, James (Texas A&M University)		
Grant/Contract No.:	NNX13AM43G		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	This research proposal is aimed at better understanding the basic bone biological effects of various osteoporosis treatments as therapies for reducing or preventing bone loss due to space flight. The project will study hindlimb unloaded (HU) rats, which is a technique that is well established as an effective and informative ground-based analog for studying the effects of space flight on the musculoskeletal system. In order to better understand the detailed effects of each, the treatments will be studied separately. The four treatments to be examined include two anabolic (bone building) therapies and two anti-resorptive therapies. One anabolic approach will use voluntary exercise and the other a new osteoporosis drug (anti-sclerostin antibody). Anti-resorptive drugs are the most popular for treating clinical osteoporosis and achieve their benefits by preventing bone resorption, which is part of the normal cycle of bone replenishment. Furthermore, the treatments will be used as pre-treatments and only administered prior to HU exposure. This is not realistic for astronauts, but it will allow more in-depth study of the fundamental nature of each treatment, and it will also align with the anticipated requirements of possible follow-up space flight experiments using the same rat animal model. The experiments to be conducted will characterize beneficial effects of pre-treatment upon bone loss due to HU as well as bone recovery following HU and return to weightbearing. Bone properties will be quantified using CT-based analyses, histological techniques for assessing bone cell function, as well as biomechanical testing to measure strength.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Completion of the studies outlined in this research proposal will provide new insights into the basic nature of the effects of exercise and pharmaceutical agents and how these interact with mechanical unloading and recovery therefrom. Removal of weightbearing through hindlimb unloading is a severe and acute form of mechanical disuse that has been shown to simulate many of the effects of microgravity exposure due to space flight on the musculoskeletal system. Hindlimb unloading also induces many musculoskeletal system changes that are similar to accelerated aging as well as other disuse scenarios such as stroke, spinal cord injury, or neuromuscular impairment. Thus, insights gained should also help advance basic understanding of the fundamental biology associated with Earth-bound clinical medicine as well as space flight effects. Future space flight experiments can be designed to replicate the research to be conducted in this proposed project, and comparing results from the two will provide essential and more detailed understanding of the full efficacy, as well as limitations, of the adult rat HU model as a ground-based analog for human space flight. Studying these treatments separately is not necessarily realistic for actual mission scenarios, but the results will nevertheless provide evidence-based rational data for contemplating and designing possible co-therapies using either combinations (simultaneously) or possible sequential scheduling protocols.
Task Progress:	The primary focus of effort during year 3 was on conducting (and completing) the second set of animal experiments. With reference to the experiments and specific aims as originally proposed (overviewed above), the order was reversed, and the anti-resorptive pre-treatment portion of the study was done first (Year 2). Accordingly, the focus for year 3 was on the anabolic Animal groups and timelines are summarized below. Anabolic Pre-Treatment Animal Study: Four experimental groups are specified for each of the two sets of animal experiments. The two anabolic pre-treatments are pharmaceutical and mechanical. Specifically, the pharmaceutical agent used was an anti-sclerostin anti-body compound provided by Lilly (Eli Lilly and Company, Indianapolis, IN) through a Material Transfer Agreement. The mechanical anabolic consists of a voluntary rat jumping protocol intended to simulate resistance exercise. The four groups for this part of the study are thus: Control (CON) ; HU Control (HUC) Sclerostin Anti-body (SAB) ; Exercise (REX) In vivo measures were made on all animals every 28 days. These consist primarily of pQCT (peripheral Quantitative Computed Tomography) scans conducted on anesthetized animals at the proximal tibia metaphysis (PTM) and at the tibia diaphysis (TD), or midshaft. In addition, blood serum samples have been acquired at each 28-day interval and stored for possible future analysis of bone activity biomarkers. A host of additional analyses are conducted ex vivo on bone specimens harvested from the animals following euthanasia at each of the time 28-day time points. Ex Vivo Endpoint Measures: Densitometry and microstructure by microCT (tibia and femur; metaphyses, midshaft, FN): ~ trabecular microarchitecture, vBMD, BMC, TMD, etc. Static and dynamic histomorphometry (tibia and femur; metaphyses, midshaft, FN): ~ trabecular, cortical, MS/BS, MAR, BFR, etc. Bone Strength (tibia metaphysis RPC, midshaft; femur midshaft, FN): ~ ultimate load, stiffness, ultimate strength, elastic modulus, toughness, etc. Muscle Weights (hindlimb posterior crural muscles): ~ wet weights for gastrocnemius, soleus, and plantaris The animal work for the anabolic pre-treatment portion of the project was originally scheduled to start in Spring of 2015, but this was delayed significantly due to acquisition of the anti-sclerostin anti-body agent and establishment of proper dosing. Considerable time and effort were expended during year 2 on this task, and a Material Transfer Agreement (MTA) was finally executed with Lilly near the very end of Year 3 (on July 24, 2015). Milestones are shown below highlighting the tasks and timing of effort for Year 3. The first several months were devoted to two pilot studies: (1) to determine proper doses for efficacy of the rat version of this drug for the strain, age, and gender (male) animals being used in this project; and (2) to continue development of jumping exercise protocol details. Both of these pilot studies were successful and laid the necessary groundwork for planning and conducting the primary animal studies. The dose suggested by Lilly (1 x per week 25mg/kg body weight) proved to be effective over 28 days of pre-treatment. The pilot studies did not include any hindlimb unloading (HU) periods; that is, they ended after 28 days of pre-treatment. The main focus of the exercise pilot study was to establish details for the minor food restriction needed to train the animals. The voluntary training protocol relies on slight food restriction in order to enhance the effectiveness of food "treats" as the positive reinforcement. The goal was to establish the amount of treats needed for effective training while also making sure the animals maintained healthy body weights. Following the pilot studies, a decision was made to to stagger the animal experiments for the drug (Scl-Ab) and for the exercise. This was based on a combination of several factors. First, this project specifies adult male Sprague-Dawley rats 5.5 to 6 months old, and the vendor requires considerable lead time in order to provide aged animals, especially considering that a total of 140-150 animals are needed. Further, the jumping exercise is very labor intensive and still needed more pilot work for additional refinement and finalizing of procedural details, particularly for establishing

	efficacy during HU exposure following pre-treatment. In addition, the logistical arrangements with Lilly necessitated acquiring and using the Scl-Ab drug early in calendar year 2016. Consequently, all of the animal work for the Scl-Ab pre-treatment was done first and in 3 cohorts. The first of these cohorts included 5 animals for finalizing the jumping exercise pre-treatment protocol and generation of proof-of-concept data. Because of the cumulative delays in conducting the anabolic pre-treatment arm of the project, a request was made and granted for a 1-year no-cost extension. The end date of the project is now August 31, 2017. Accordingly, the milestones for Year 3 can be summarized as follows: Major Milestones for Year 3 Oct. 2016: Receipt of initial shipment of Scl-Ab Nov. – Dec. 2016: Pilot study for Scl-Ab dosing; Pilot study for voluntary jumping exercise Jan. – Sep. 2016: Animal work for Scl-Ab drug pre-treatment conducted in 3 cohorts-- Cohort 1 – 26 Jan 2016 to 22 June 2016 (20 Scl-Ab; 5 REX); Cohort 2 – 18 Apr 2016 to 31 Aug 2016 (13 Scl-Ab); Cohort 3 – 5 May 2016 to 14 Sept 2016 (12 Scl-Ab) May 2016: No-Cost Extension Granted (1 year) Throughout the duration of year 3 progress was also maintained on ex vivo measures and analyses for the first part of the project, the anti-resorptive drug pre-treatment arm.
Bibliography Type:	Description: (Last Updated: 01/11/2021)
Abstracts for Journals and Proceedings	Brezicha JE, Lenfest SE, Koskniewski JL, Leach C, Black JM, Allen MR, Bloomfield SA, Hogan HA. "Pre-Treatment with Bisphosphonates Mitigates Bone Loss at the Tibia Metaphysis and Femoral Neck During Subsequent Hindlimb Unloading and Recovery. " Presented at 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. 2016 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 8-11, 2016. Poster #7208. , Feb-2016
Abstracts for Journals and Proceedings	Brezicha JE, Black JM, Metzger CE, Lenfest SE, Koskniewski JL, Bloomfield SA, Allen MR, Hogan HA. "Bisphosphonate Pre-Treatments Enhance Trabecular Bone Architecture during Unloading and Reambulation Despite Lower Resorption and Formation." Presented at ASBMR 2016 (American Society for Bone and Mineral Research), Atlanta, Georgia, September 16-19, 2016. ASBMR 2016 (American Society for Bone and Mineral Research), Atlanta, Georgia, September 16-19, 2016. Poster Presentation FR0023 & SA0023. , Sep-2016
Abstracts for Journals and Proceedings	Lenfest SE, Koskniewski JL, Looper AL, Brezicha JE, Black JM, Bloomfield SA, Fluckey JD, Hogan HA. "Initial Results for a New Positive Reinforcement Voluntary Jumping Exercise in Rats Show Enhanced Bone Parameters." Presented at ASBMR 2016 (American Society for Bone and Mineral Research), Atlanta, Georgia, September 16-19, 2016. ASBMR 2016 (American Society for Bone and Mineral Research), Atlanta, Georgia, September 16-19, 2016. Poster Presentation MO0030. , Sep-2016